

Seyedarmin (Armin) Azizi

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RESEARCH INTERESTS

Efficient AI, post-training and alignment, foundation models, LLM reasoning and inference, reinforcement learning, optimization.

EDUCATION

University of Southern California (USC) Los Angeles, CA

Ph.D. in Electrical and Computer Engineering Aug. 2022 – Present

Advisor: Prof. Massoud Pedram. GPA: 4.0/4.0.

Dissertation direction: post-training methods and efficient inference for large-scale AI models.

University of Southern California (USC) Los Angeles, CA

M.S. in Electrical Engineering. GPA: 4.0/4.0 2022 – 2024

Sharif University of Technology (SUT) Tehran

B.Sc. in Electrical Engineering, Digital Systems. GPA: 18.6/20.0 (3.84/4.0) 2017 – 2021

RESEARCH EXPERIENCE

Apple Inc., SIML Team — Applied Research Intern Seattle, WA · Summer 2026

- Traced response collapse in instruction-tuned LLMs to reward hacking under rubric-based RL.
- Built a benchmark that measures the failure, and proposed a mechanistically grounded mitigation at inference time.
- Designed an improved DPO and rubric-based RL recipe that removes the cause upstream.
- Presented the results to Apple AIML leadership.

USC SPORT Lab — Graduate Research Assistant Los Angeles, CA · 2022 – Present

- Develop post-training and inference-time methods for LLM reasoning, which include activation steering, low-rank steering, and low-dimensional adaptation for deployment on resource-constrained devices.
- Design representation-level interventions that decrease chain-of-thought cost without a loss of accuracy.
- Extend long-context generalization for Transformer and state-space models, with evaluation on retrieval, summarization, and healthcare tasks.

Semiconductor Research Corporation (SRC) Project, USC 2023 – 2024

- Built a mixed-computation (posit and floating-point) inference framework for DNNs on edge devices.
- Designed the model-to-format mapping procedure, the customized training flow, and the MAC unit kernel. The system improves accuracy and efficiency against an IEEE-754 baseline.

Intel Labs — Research Collaboration Mentor: Dr. Souvik Kundu · 2024 – Present

- Developed long-context extension methods for Transformer-based and Mamba-based language models. The work was published at EMNLP 2024 (Findings) and ICLR 2025.

Sharif University of Technology — Undergraduate Research Tehran, Iran · 2020 – 2021

- Built a YOLOv5-based vehicle detection model for traffic applications, which included data collection, annotation, training, and post-processing.
- Developed object tracking and video synopsis methods with YOLO and DeepSORT for video summarization.

SELECTED PUBLICATIONS

* denotes equal contribution.

Conference and Journal Papers

1. J. Tian, **S. Azizi**, Y. Zhao, E. Baghaei, et al., "SkipKV: Selective Skipping of KV Generation and Storage for Efficient Inference with Large Reasoning Models," *MLSys 2026*. **Oral**.
2. **S. Azizi**, E. Baghaei, S. Kundu, M. Pedram, "Activation Steering for Chain-of-Thought Compression," *ACL 2026*.
3. **S. Azizi***, E. Baghaei*, S. Kundu, M. Pedram, "Top-H Decoding: Adapting the Creativity and Coherence with Bounded Entropy in Text Generation," *NeurIPS 2025*.
4. **S. Azizi**, S. Kundu, M. Sadeghi, M. Pedram, "MambaExtend: A Training-Free Approach to Improve Long Context Extension of Mamba," *ICLR 2025*.
5. **S. Azizi**, S. Kundu, M. Pedram, "LaMDA: Large Model Fine-Tuning via Spectrally Decomposed Low-Dimensional Adaptation," *EMNLP 2024*.
6. M. Sadeghi, A. Fayyazi, **S. Azizi**, M. Pedram, "PEANO-ViT: Power-Efficient Approximations of Non-Linearities in Vision Transformers," *ISLPED 2024*.
7. **S. Azizi**, M. Nazemi, A. Fayyazi, M. Pedram, "Automated Optimization of Deep Neural Networks: Dynamic Bit-Width and Layer-Width Selection via Cluster-Based Parzen Estimation," *DATE 2024*.
8. **S. Azizi**, R. Soleimani, M. Ahmadi, et al., "Performance Enhancement of an Uncertain Nonlinear Medical Robot," *Computers in Biology and Medicine (Elsevier)*, 2022.

Preprints and Papers Under Review

9. **S. Azizi**, et al., "Power-SMC: Low-Latency Sequence-Level Power Sampling for Training-Free LLM Reasoning," under review, NeurIPS 2026. arXiv:2602.10273.
10. E. Baghaei*, **S. Azizi***, et al., "One Token Away from Collapse: The Fragility of Instruction-Tuned Helpfulness," under review, NeurIPS 2026. arXiv:2604.13006.

Workshop Papers

11. A. Aryashad, et al., "From Filters to VLMs: Benchmarking Defogging Methods through Object Detection and Segmentation Performance," *WACV 2026*. **Oral**.
12. T. Khezresmaeilzadeh, P. Razmara, **S. Azizi**, M. E. Sadeghi, E. Baghaei, "VISTA: Vision-Language Inference for Training-Free Stock Time-Series Analysis," *CVPR 2025 Workshop on Transformers for Vision*.
13. **S. Azizi**, S. Kundu, M. Sadeghi, M. Pedram, "QMambaExtend: Improving Long-Context Extension of Memory-Efficient Mamba Models," *ICLR 2025 Workshop on Scalable Optimization for Efficient and Adaptive Foundation Models*.
14. **S. Azizi**, M. Nazemi, M. Pedram, "Memory-Efficient Vision Transformers: An Activation-Aware Mixed-Rank Strategy," *ECCV 2024 Workshop on Efficient Foundation Models*.
15. J. Heo, **S. Azizi**, A. Fayyazi, M. Pedram, "Training-Free ViT Acceleration with Delayed Spatial Merging," *ICML 2024 ES-FoMO-II Workshop*.

HONORS AND AWARDS

- 2nd place, System Design Contest, Design Automation Conference (DAC) 2024, among 31 international teams.
- Ranked 1st, ECE Ph.D. Screening Exam, USC.
- ACM Complimentary Membership Award, 2024.
- DAC Young Fellow Award, 2023.
- Highest GPA in the Digital Systems track, Sharif University of Technology.
- Ranked 9th out of 60,000 participants, Iran National University Entrance Exam.

PROFESSIONAL SERVICE AND TEACHING

- Reviewer: ICLR, NeurIPS, ICML, ACL, ECCV, WACV.
- Teaching Assistant: Machine Learning, ML Hardware Acceleration, Logic Circuits, Digital Systems.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, CUDA, MATLAB, Bash, Verilog, VHDL, Java, OpenCL.
- **ML frameworks:** PyTorch, JAX, TensorFlow, Hugging Face (Transformers, PEFT, Diffusers), DeepSpeed, Megatron-LM, vLLM, SGLang, scikit-learn, OpenCV, timm.
- **Systems and hardware:** Linux, Git, gcc/make, LLVM, MLIR, HLS, Vivado, Quartus II, Xilinx ISE.
- **Data and cloud:** SQL, R, AWS (Bedrock, S3, API Gateway), Azure.